FDT-N: Footprint Distribution Transformer–Narrow

Inputs, Architecture, and Direct Multi-Horizon Quantile Outputs

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Abstract

Volume-footprint bars are variable-size sets of executed-volume rows. Footprint Distribution Transformer–Narrow (FDT-N) maps a 64-bar ragged history to direct one- through eight-minute marginal quantile forecasts. Its exported graph accepts four tensors: 11 features per price row, a row mask, 23 engineered features per bar, and a bar mask. Independent row embeddings feed four-query masked pooling; a gate combines each row-set summary with its aggregate token; three causal temporal blocks encode the history; and two decoder blocks process eight learned horizon queries. The 371,861-parameter model returns $[B, 8, 7, 3]$: batch, horizon, target, and q10/q50/q90. Targets are total POC, low, high, buy POC, sell POC, relative log volume, and delta fraction. Softplus spreads order the marginal quantiles; deterministic maps impose price-range relations and bound delta fraction.

The development corpus contains 20,160 one-minute BTCUSDT spot bars, 357,676 accepted rows, and 20,089 continuous 64+8 windows from January 1–14, 2024. A purged chronological split yields 13,990 training, 2,869 validation, and 2,942 test windows. Across three seeds, test pinball loss is 0.07039–0.07154 and q50 mean absolute error is 0.23587–0.23724. For the selected seed, q50 error is 0.235870, versus 0.254298 for causal persistence and 0.229719 for train-only direct ridge; central-80% marginal coverage is 78.98%. PyTorch and ONNX Runtime agree to maximum absolute error 2.3842×10^{-7} on the artifact-bound example.

Keywords: volume footprint; ragged sets; transformer; marginal quantile regression; multi-horizon forecasting; market microstructure.

1 Introduction

A volume-footprint bar records executed buy- and sell-initiated volume at each traded price. Unlike a conventional open-high-low-close-volume vector, the number of rows varies with the realized price range. A model must therefore combine two structures: an order-invariant set stored in canonical order and an ordered sequence of completed bars. Collapsing the rows to a few aggregates is computationally convenient, but may erase distributional shape. Padding all bars to a large fixed grid preserves more detail but wastes computation and makes overflow behavior easy to hide.

FDT-N is a small hierarchical transformer designed around this representation mismatch. It uses learned queries to reduce each variable-size row set to a fixed-width token and then models 64 completed-history tokens with a triangular attention mask. A parallel decoder produces eight horizons without feeding earlier forecasts back into later ones. The output layer enforces within-target quantile order and elementary price relationships. The final trained model contains 371,861 parameters.

"Narrow" names this fixed compact configuration: width 64, four attention heads, three temporal blocks, two decoder blocks, and 371,861 trainable parameters. It distinguishes the reported artifact from a broader family of possible footprint transformers.

This paper answers two implementation questions directly: what tensors enter FDT-N, and what tensor does it return? It then reports a chronological development benchmark against causal persistence and train-only direct ridge. The selected FDT-N q50 has lower pooled error than persistence, while ridge has the lowest pooled q50 error in this comparison. Those measurements characterize this corpus and split; the architectural specification stands independently of comparator rank.

This paper makes four bounded contributions:

1. It specifies the four-input ONNX contract, tensor shapes, axis order, dtypes, dynamic axes, chronological order, preprocessing boundary, and explicit 1,024-row acceptance bound.
2. It derives learned-query row pooling, gated fusion, cutoff-causal temporal processing, the parallel horizon decoder, ordered marginal quantiles, physical constraints, and weighted pinball objective.
3. It reports a hash-bound 14-day development experiment with three seeds, causal persistence, train-only direct ridge, per-target and per-horizon results, descriptive paired block resampling, export parity, and CPU timing.
4. It traces one artifact-bound window from the four input tensors through named intermediate tensors to normalized and physical output values, with PyTorch–ONNX parity.

2 FDT-N at a glance: inputs, processing, and outputs

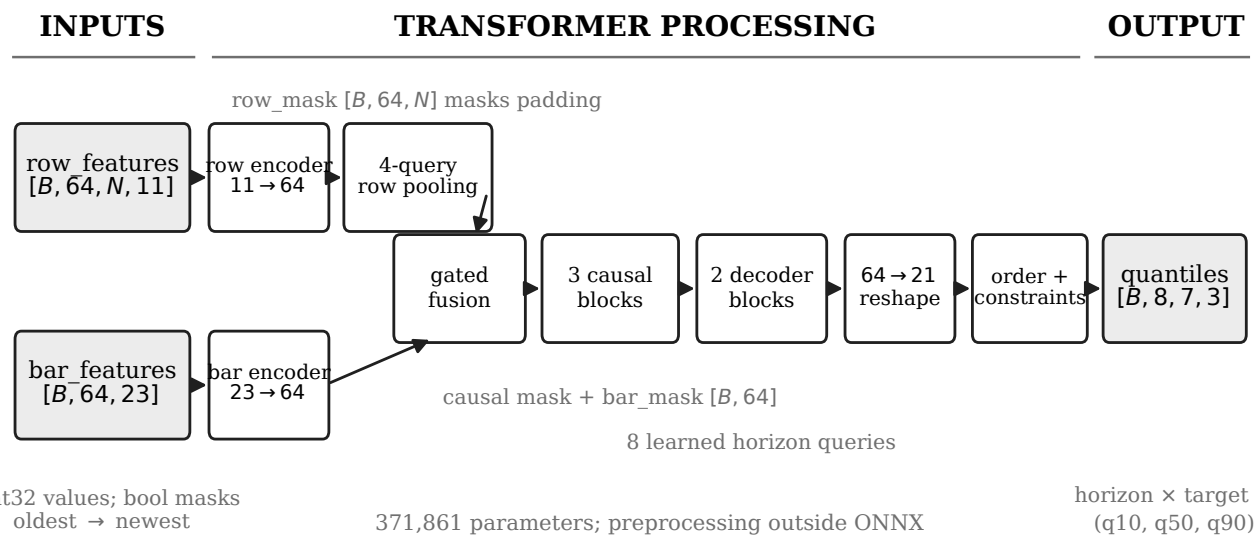


Figure 1: The complete FDT-N interface and forward path. Preprocessing produces four input tensors; masked row pooling and temporal/decoder attention transform them; ordered and physically constrained marginal quantiles form the single output tensor.

Table 1: Exported ONNX interface. B is dynamic batch size and N is the dynamic batch-local row width. Time runs oldest to newest. Preprocessing and inverse physical transforms are outside ONNX.

Name	Type	Shape	Meaning / axis order
`row_features`	float32	$[B, 64, N, 11]$	eleven encoded features for every stored row
`row_mask`	bool	$[B, 64, N]$	true exactly where a row is observed
`bar_features`	float32	$[B, 64, 23]$	engineered bar/context features per completed bar
`bar_mask`	bool	$[B, 64]$	true exactly where a history token is valid
`quantiles`	float32	$[B, 8, 7, 3]$	horizon, target, quantile; target order POC/low/high/buy-POC/sell-POC/log-volume/delta and quantile order .1/.5/.9

For a batch of histories, the implemented forward map is

$$(X, M_R, F, M_B) \longmapsto E \longmapsto A \longmapsto U \longmapsto H \longmapsto D \longmapsto R \longmapsto Q, \tag{1}$$

with shapes

$$\begin{aligned} &X : [B, 64, N, 11], \quad && E : [B, 64, N, 64], \quad && A : [B, 64, 64], \\ &F : [B, 64, 23], && U, H : [B, 64, 64], && D : [B, 8, 64], \\ &R : [B, 8, 7, 3], && Q : [B, 8, 7, 3]. \end{aligned} \tag{2}$$

M_R masks padded rows and M_B masks padded history positions. R stores the raw (c, s^-, s^+) head values; softplus ordering and target constraints convert it to Q. The returned price and log-volume values are continuous normalized coordinates and may extend beyond the clipped label range. Price inverse transforms can therefore lie between footprint-grid rows unless a consumer applies a separately specified snapping policy. Buy- and sell-POC outputs are conditional on the corresponding side being present in training labels; FDT-N emits no side-presence probability.

Table 2: Final architecture and exact trainable-parameter accounting.

Component / property	Value
Row encoder, four-query pooling, row summary	63,488
Bar projection and gated residual fusion	22,272
Three temporal encoder blocks	166,784
Two-block eight-query horizon decoder	117,888
Output RMSNorm and $64 \to 21$ head	1,429
Total trainable parameters	371,861
Model width / heads / head width	64 / 4 / 16
Dropout / initial LayerScale	0.10 / 10^{-2}

3 Related work

3.1 Sets, attention, and ragged observations

Deep Sets establishes a general permutation-invariant form based on elementwise representation and symmetric aggregation [23]. Set Transformer replaces fixed summation with attention-based interactions and learned inducing or seed vectors [11]. Perceiver similarly uses a small latent array to query large inputs [6]. FDT-N adopts the narrowest transferable idea: a constant number of learned queries cross-attends to the observed rows. It does not use row-to-row self-attention, iterative latent refinement, or a claim that attention weights explain market behavior [7].

3.2 Multi-horizon forecasting

Temporal Fusion Transformer combines recurrent processing, gating, attention, and covariate selection for multi-horizon probabilistic prediction [12]. PatchTST tokenizes temporal patches [15]; iTransformer treats variables as tokens [13]; and Moirai studies universal time-series pretraining [22]. DLinear demonstrates that simple linear models can be informative forecasting controls [24]. Accordingly, direct ridge supplies a scale reference for FDT-N's q50 output. FDT-N is a task-specific 64-token model rather than a foundation model.

3.3 Market microstructure

DeepLOB applies convolutional and recurrent layers to fixed-depth limit-order-book states [26]. FI-2010 provides a benchmark for mid-price prediction from limit-order-book data [16]; later work studies transformers and multi-horizon order-flow imbalance [9, 21]. A footprint is different from a book: it summarizes executed trades by price inside a completed interval, whereas a limit-order-book tensor describes displayed resting liquidity. Results do not transfer automatically between the two.

3.4 Probabilistic prediction under dependence

Quantile regression uses asymmetric absolute loss to estimate conditional quantiles [8], and proper scoring rules provide a principled basis for probabilistic evaluation [4]. Noncrossing constraints can be estimated or imposed by parameterization [2]. Conformalized quantile regression can adjust intervals under a separate calibration design and stated assumptions [17]; this study has no such partition. Because adjacent forecast windows overlap, we use paired contiguous block resampling inspired by dependent-data bootstrap methods [10]. We do not present it as an independent-block confidence interval or a Diebold–Mariano test [3].

4 Volume-footprint representation

4.1 Executed-volume rows

For price row j in completed bar t, define

$$r_{t,j} = \left(p_{t,j}, v^{+}_{t,j}, v^{-}_{t,j}, I^{bid}_{t,j}, I^{ask}_{t,j}, z^{bid}_{t,j}, z^{ask}_{t,j}\right), \tag{3}$$

where p is the snapped row price, v^{+} is buy-initiated (ask-lifting) volume, v^{-} is sell-initiated (bid-hitting) volume, I are diagonal-imbalance flags, and z are explicit zero-side flags. Total executed volume and signed delta at the row are

$$v_{t,j} = v^{+}_{t,j} + v^{-}_{t,j}, \qquad \Delta_{t,j} = v^{+}_{t,j} - v^{-}_{t,j}. \tag{4}$$

Adding and subtracting these two equations gives the inverse identities

$$v^+_{t,j} = \frac{v_{t,j} + \Delta_{t,j}}{2}, \qquad v^-_{t,j} = \frac{v_{t,j} - \Delta_{t,j}}{2}. \tag{5}$$

Since both side volumes are nonnegative, $v + \Delta \geq 0$ and $v - \Delta \geq 0$; therefore

$$-v_{t,j} \leq \Delta_{t,j} \leq v_{t,j}, \qquad |\Delta_{t,j}| \leq v_{t,j}. \tag{6}$$

This is an accounting constraint, not a directional trading rule. Positive delta only says that aggressive buy volume exceeded aggressive sell volume at that row.

For imbalance ratio λ (3 by default), the stored diagonal flags compare opposite sides at adjacent prices:

$$I^{bid}_{t,j} = \mathbf{1}\left[v^-_{t,j} > 0 \ \wedge \ v^-_{t,j} \geq \lambda v^+_{t,j-1}\right], \qquad I^{ask}_{t,j} = \mathbf{1}\left[v^+_{t,j} > 0 \ \wedge \ v^+_{t,j} \geq \lambda v^-_{t,j+1}\right]. \tag{7}$$

The exact neighbor direction follows the high-to-low row ordering; a missing outer neighbor has volume zero. The flags record a thresholded relationship; they do not preserve the underlying ratio.

4.2 From rows to bar statistics

Let $R_t = \{r_{t,1}, \ldots, r_{t,n_t}\}$ contain every accepted row of bar t. Side, total, and delta volumes are additive:

$$V^+_t = \sum_{j=1}^{n_t} v^+_{t,j}, \quad V^-_t = \sum_{j=1}^{n_t} v^-_{t,j}, \quad V_t = V^+_t + V^-_t, \quad \Delta_t = V^+_t - V^-_t. \tag{8}$$

The point of control (POC) is a mode, not an average:

$$j^*_t = \arg\max_j v_{t,j}, \qquad p^{POC}_t = p_{t,j^*_t}. \tag{9}$$

Buy and sell POCs replace $v_{t,j}$ in Eq. (9) by $v^+_{t,j}$ and $v^-_{t,j}$. Rows are sorted from high to low, and ties resolve to the first maximum, hence the highest price. For an all-zero side, this argmax still returns the highest observed row, but the corresponding future side-POC target is masked. By contrast, the volume centroids are weighted means:

$$c_t = \frac{\sum_j p_{t,j} v_{t,j}}{V_t}, \quad c^+_t = \frac{\sum_j p_{t,j} v^+_{t,j}}{V^+_t} \ (V^+_t > 0), \quad c^-_t = \frac{\sum_j p_{t,j} v^-_{t,j}}{V^-_t} \ (V^-_t > 0). \tag{10}$$

A POC answers "where was the most volume at one row?"; a centroid answers "where was the average unit of volume?" They coincide only in special cases. A side centroid is mathematically undefined when that side has zero volume; the implemented FDT preprocessing fills its fixed input slot with the total-POC price. The future side-POC target is masked whenever its side volume is zero.

For target fraction $\eta = 0.70$, the value area is a deterministic contiguous interval around the total POC. Let $[\ell_m, u_m]$ be the current high-to-low index interval, initialized at $\ell_0 = u_0 = j^*_t$. Define the available volume immediately above and below as

$$a_m = \begin{cases} v_{t,\ell_m - 1}, & \ell_m > 1, \\ -1, & \text{otherwise}, \end{cases} \qquad b_m = \begin{cases} v_{t,u_m + 1}, & u_m < n_t, \\ -1, & \text{otherwise}. \end{cases} \tag{11}$$

If $b_m > a_m$, set $u_{m+1} = u_m + 1$; otherwise set $\ell_{m+1} = \ell_m - 1$, leaving the other bound unchanged. Thus an equal-volume tie expands upward to the higher price. Continue until

$$\frac{\sum_{j=\ell_m}^{u_m} v_{t,j}}{V_t} \geq \eta. \tag{12}$$

The final lowest and highest prices are value-area low and high. Contiguity makes this different from selecting the globally largest 70% of rows. Bar high and low are the extreme observed row prices. Cumulative volume delta obeys

$$CVD_t = CVD_{t-1} + \Delta_t, \tag{13}$$

so its reset policy is part of the coordinate definition. In this corpus, CVD starts at zero at the beginning of the 14-day build and carries across sealed bars. Feature 15 later subtracts the first context CVD, making it invariant to an additive global reset within the encoded series.

Causal availability

A bar is usable only after its interval has ended and the configured feed-completeness decision is available. No event, normalization statistic, model-selection choice, or calibration value whose availability follows the forecast cutoff may affect the input. Event time alone is insufficient when late data can arrive.

5 Input construction and forecasting task

5.1 Information set and direct horizons

Let $B_t = (R_t, b_t)$ denote the row set and engineered bar/context features of completed bar t. For lookback $L = 64$ and horizon count $H = 8$, the forecast map is

$$\widehat{Y}_{t+1:t+H} = F_\theta(B_{t-L+1:t}), \qquad \widehat{Y} \in \mathbb{R}^{H\times K\times Q}, \tag{14}$$

with $K = 7$ targets and $Q = 3$ quantile levels $\mathcal{T} = \{0.1, 0.5, 0.9\}$. The information set ends at the seal of B_t. Every adjacent timestamp across the 64 input bars and 8 label bars must differ by exactly 60 seconds. Thus a candidate starting at s is valid only if

$$T_{i+1} - T_i = 60 \text{ seconds} \quad \text{for every } i = s, \dots, s + L + H - 2. \tag{15}$$

Missing or irregular intervals invalidate the window; they are not treated as adjacent market time.

The model is direct: all eight outputs are functions of the same historical information set,

$$(\widehat{Y}_1, \dots, \widehat{Y}_8) = F_\theta(B_{t-63:t}), \tag{16}$$

rather than an autoregression $\widehat{Y}_h = f(B_{t-63:t}, \widehat{Y}_{1:h-1})$. This distinction removes forecast-as-input error propagation but does not make errors across horizons independent.

5.2 Ragged storage and masks

Row counts n_t vary by bar. The encoded series therefore concatenates rows into one matrix $E \in \mathbb{R}^{N\times 11}$ and stores offsets $o_0 = 0$ and

$$o_{t+1} = o_t + n_t, \qquad R_t \longleftrightarrow E[o_t : o_{t+1}, :]. \tag{17}$$

Within a window, shorter bars are padded to the widest observed local row count; batching may then pad windows to the batch-local maximum. A binary mask $m_{t,j}$ equals 1 for observed rows and 0 for padding. In attention logits, padding receives $-\infty$, so its softmax weight is exactly zero in ideal arithmetic. Batches are limited by both item count and padded row-width budget; wide windows produce smaller batches rather than dropped rows. The export keeps the row axis dynamic.

The accepted-row contract is $n_t \le 1{,}024$; a bar outside that domain is not encoded. Every row is retained for accepted bars. In the frozen corpus, the maximum is 605.

5.3 Input coordinates

Let $\delta_p = 2.5$ USDT be the footprint-row size and let $\kappa(p) = p/\delta_p$ denote the integer row coordinate after snapping. Accepted bars satisfy $V_t > 0$. For each row, define $S_g = 32$ row increments. The implemented 11-vector is

$$x^{(1)}_{t,j} = \operatorname{clip}\left(\frac{\kappa(p_{t,j}) - \kappa(p^{POC}_t)}{S_g}, -2, 2\right), \tag{18}$$

$$x^{(2:4)}_{t,j} = \left(\frac{v^+_{t,j}}{V_t}, \frac{v^-_{t,j}}{V_t}, \frac{v_{t,j}}{V_t}\right), \tag{19}$$

$$x^{(5)}_{t,j} = \frac{\Delta_{t,j}}{V_t}, \tag{20}$$

$$x^{(6:7)}_{t,j} = \left(\frac{\log(1+v^+_{t,j})}{\log(1+V_t)}, \frac{\log(1+v^-_{t,j})}{\log(1+V_t)}\right), \tag{21}$$

$$x^{(8:11)}_{t,j} = (I^{bid}_{t,j}, I^{ask}_{t,j}, z^{bid}_{t,j}, z^{ask}_{t,j}). \tag{22}$$

When $V_t > 0$, the side shares sum to the total share, and summing the total share across rows gives 1. The logarithmic ratios compress scale while retaining zeros. The binary flags remain discrete facts.

The 23-dimensional bar vector is separately projected. Table 3 states its schema order; reordering it would change the model coordinate.

Table 3: Implemented aggregate bar features in schema order. All price quantities are normalized row-coordinate offsets; references are computed from context only.

Index	Feature(s)	Meaning
1	POC offset	total POC relative to the last-context anchor POC
2–3	buy/sell POC offsets	side POCs relative to the bar POC
4–6	total/buy/sell centroid offsets	Eq. (10) relative to the bar POC
7–10	low, high, value-area gaps	lower and upper structure around the bar POC
11	relative log volume	log volume relative to a context-only median
12–14	buy, sell, delta fractions	V^+/V, V^-/V, and Δ/V
15	cumulative-delta change	context-relative CVD change, scaled by 4
16–17	stacked buy/sell fractions	normalized run-length summaries
18	row-count fraction	log-scaled row count under the 1,024-row bound
19	feed quality	raw source-quality scalar (1.0 in this corpus)
20–23	calendar cycle	sine/cosine of minute-of-day and weekday

The exact aggregate transform follows. Express prices as row ticks and write, for context bar r, P_r for total POC; P_r^+, P_r^- for side POCs; C_r, C_r^+, C_r^- for centroids; L_r, H_r for low/high; A_r^L, A_r^H for value-area bounds; D_r for CVD; S_r^+, S_r^- for stacked-run lengths; and n_r for row count. The context-only references are

$$P^* = P_t, \qquad \widetilde{\ell}_t = \text{median}\{\log(1+V_r) : r = t-15, \ldots, t\}, \qquad V^* = \exp(\widetilde{\ell}_t) - 1, \qquad D_0 = D_{t-L+1}. \tag{23}$$

For $S_p = 64$, $S_g = 32$, $S_v = S_c = 4$, and $n_{max} = 1024$, features 1–15 are

$$f_{r,1} = \text{clip}((P_r - P^*)/S_p, -2, 2), \tag{24}$$

$$(f_{r,2}, f_{r,3}) = \big(\text{clip}((P_r^+ - P_r)/S_g, -2, 2), \text{clip}((P_r^- - P_r)/S_g, -2, 2)\big), \tag{25}$$

$$(f_{r,4}, f_{r,5}, f_{r,6}) = \big(\text{clip}((C_r - P_r)/S_g, -2, 2), \text{clip}((C_r^+ - P_r)/S_g, -2, 2), \text{clip}((C_r^- - P_r)/S_g, -2, 2)\big), \tag{26}$$

$$(f_{r,7}, f_{r,8}) = \big(\min((P_r - L_r)/S_g, 2), \min((H_r - P_r)/S_g, 2)\big), \tag{27}$$

$$(f_{r,9}, f_{r,10}) = \big(\min((P_r - A_r^L)/S_g, 2), \min((A_r^H - P_r)/S_g, 2)\big), \tag{28}$$

$$f_{r,11} = \text{clip}((\log(1+V_r) - \widetilde{\ell}_t)/S_v, -2, 2), \tag{29}$$

$$(f_{r,12}, f_{r,13}, f_{r,14}) = \left(V_r^+/V_r, V_r^-/V_r, (V_r^+ - V_r^-)/V_r\right), \tag{30}$$

$$f_{r,15} = \text{clip}\left(\frac{D_r - D_0}{S_c \max(V^* L, 1)}, -2, 2\right). \tag{31}$$

The structural gaps in features 7–10 are expected nonnegative, so the implementation caps only the upper side. The remaining features are

$$f_{r,16} = \min(S_r^+ / \max(n_r, 1), 1), \qquad f_{r,17} = \min(S_r^- / \max(n_r, 1), 1), \tag{32}$$

$$f_{r,18} = \min\left(\frac{\log(1+n_r)}{\log(1+n_{max})}, 1\right), \qquad f_{r,19} = q_r, \tag{33}$$

$$f_{r,20:21} = (\sin(2\pi m_r/1440), \cos(2\pi m_r/1440)), \tag{34}$$

$$f_{r,22:23} = (\sin(2\pi w_r/7), \cos(2\pi w_r/7)). \tag{35}$$

For bar-start time T_r in Unix milliseconds, $m_r = (\lfloor T_r/60000 \rfloor \bmod 1440)$ and $w_r = (\lfloor T_r/86400000 \rfloor + 3) \bmod 7$, with Monday equal to zero. q_r is the raw quality field; it is 1.0 throughout this frozen Binance corpus. Every reference in Eqs. (23)–(31) is constructed only from archive events assigned to bars ending by the final event-time cutoff, even though an early token uses context-wide reference constants as discussed in Section 6.4. Section 11 summarizes the distinction between event-time reconstruction and live receive-time availability.

5.4 Targets and inverse transforms

Let $\mathcal{P} = \{POC, low, high, buyPOC, sellPOC\}$, $S_p = 64$ row increments, and $S_v = 4$. For future horizon h and price target $k \in \mathcal{P}$,

$$y_h^k = \text{clip}\left(\frac{\kappa(p_{t+h}^k) - \kappa(p_t^{POC})}{S_p}, -2, 2\right). \tag{36}$$

Low, high, and side POCs are direct offsets from the cutoff anchor; they are not gaps from an unknown future POC. The remaining targets are

$$y_h^V = \mathrm{clip}\left(\frac{\log(1+V_{t+h}) - \widetilde{\ell}_t}{S_v}, -2, 2\right), \tag{37}$$

$$y_h^\Delta = \mathrm{clip}\left(\frac{V_{t+h}^+ - V_{t+h}^-}{V_{t+h}^+ + V_{t+h}^-}, -1, 1\right). \tag{38}$$

Buy- and sell-POC labels are masked independently if the relevant side volume is absent.

Ignoring clipping for the moment, a normalized price quantile q_h^k maps back to physical price as

$$\widehat{p}_{t+h,\tau}^k = p_t^{POC} + \delta_p S_p \widehat{q}_{h,k,\tau}, \tag{39}$$

and a normalized volume quantile maps to

$$\widehat{V}_{t+h,\tau} = \exp\left(\widetilde{\ell}_t + S_v \widehat{q}_{h,V,\tau}\right) - 1. \tag{40}$$

The volume inverse is nonnegative exactly when $\widetilde{\ell}_t + S_v \widehat{q}_{h,V,\tau} \geq 0$. Price and log-volume heads are continuous and unbounded even though training labels are clipped; delta fraction is bounded by tanh. Consumers can retain normalized coordinates or define a domain policy for physical values. Each inverse is marginal: applying it separately to the seven targets does not construct a joint future candle.

5.5 Frozen development corpus and split

The data comprise official Binance Data Vision BTCUSDT spot individual-trade archives [1] for January 1–14, 2024. For trade event time T in milliseconds, bar start is $T - (T \bmod 60000)$. A positive price is snapped to tick $\lfloor p/2.5 + 0.5 \rfloor$. A trade is buy-initiated when Binance's `buyer_is_maker` flag is false and sell-initiated when it is true. Quantity is multiplied by 1,000, rounded with Python's ties-to-even `round` semantics, converted to integer, and dropped when the result is nonpositive. The same coordinate is used by the extractor and model package.

Table 4: Frozen development-corpus and chronological-split facts.

Item	Value
Venue / instrument / source	Binance spot BTCUSDT individual trades
UTC span	2024-01-01 00:00 through 2024-01-14 23:59
Bar interval / footprint row size	60 seconds / 2.5 USDT
Completed bars / complete accepted rows	20,160 / 357,676
Rows per bar, p50 / p95 / p99 / maximum	13 / 47 / 94.41 / 605
Valid continuous 64+8 windows	20,089
Train / validation / test windows	13,990 / 2,869 / 2,942
Purge around each split boundary	72 window starts per side; 144 per boundary
Test-context span	2024-01-12 22:50 through 2024-01-14 23:51 UTC
Calibration partition / untouched outer holdout	none / none

Table 4 fixes the venue, period, row grid, corpus size, and split counts used below. The archives support event-time reconstruction but do not preserve the original live receive time, availability decision, or late-arrival acceptance epoch. Accordingly, Eq. (15) and the split establish mechanical event-time cutoff separation, not a verified historical simulation of when every trade would have been available online. The submission bundle's lineage record begins at the official archives and preserves the encoded counts. These statements characterize data lineage rather than forecast output.

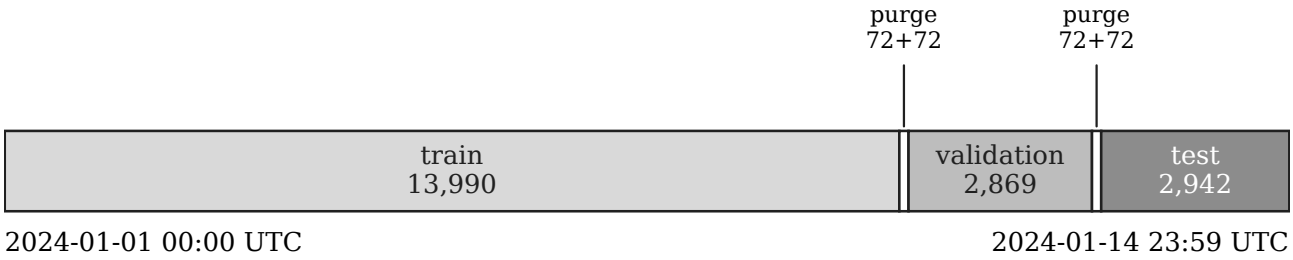


Figure 2: Chronological development split. At each boundary, 72 window starts are removed from each adjacent partition, giving 144 removed starts per boundary and 288 in total. The figure widths are proportional to the 20,089 valid window starts.

Figure 2 visualizes the chronological partitions and the two purged boundaries. A window consumes L context bars and H label bars, a span of $L + H = 72$ bar indices. Without a purge, a training label interval could overlap a validation context or label interval. The split removes 72 candidate window starts on both sides of each of the two boundaries: $4 \times 72 = 288$ starts. The identity $13{,}990 + 2{,}869 + 2{,}942 + 288 = 20{,}089$ reconciles the reported counts. This conservative separation leaves serial dependence within each partition; a later outer period is part of the follow-up design in Section 10.3.

6 Transformer processing pipeline

Figure 1 and Eq. (2) give the complete path. This section defines each learned transformation, including normalization, residual paths, masks, and dimensions.

6.1 Independent row encoder

For a bar with n_t rows, stack the features from Eq. (22) into $X_t \in \mathbb{R}^{n_t \times 11}$. The same two-layer map is applied independently to each row:

$$e_{t,j} = \mathrm{RMSNorm}(W_2\ \mathrm{GELU}(W_1 x_{t,j} + b_1) + b_2) \in \mathbb{R}^d. \tag{41}$$

Here $W_1 \in \mathbb{R}^{64\times 11}$ and $W_2 \in \mathbb{R}^{64\times 64}$, so the width is $11 \to 64 \to 64$. GELU is $z\Phi(z)$ [5].

For $u \in \mathbb{R}^d$, RMSNorm is

$$\mathrm{RMSNorm}(u) = g \odot \frac{u}{\sqrt{d^{-1}\sum_{i=1}^{d} u_i^2 + \epsilon}}, \tag{42}$$

with learned scale g [25]. RMSNorm rescales without mean subtraction. At this stage no row has interacted with another row; mixing occurs only in query pooling.

6.2 Learned-query multi-head row pooling

Let $K_q = 4$ learned query vectors form $Q_0 \in \mathbb{R}^{K_q \times d}$. The implemented cross-attention block first normalizes queries and row memory independently:

$$\bar{Q}_0 = \mathrm{RMSNorm}_Q(Q_0), \qquad \bar{E}_t = \mathrm{RMSNorm}_M(E_t), \tag{43}$$

$$C_t = Q_0 + \mathrm{Drop}_{.1}\big(\mathrm{MHA}(\bar{Q}_0, \bar{E}_t, \bar{E}_t; M_t)\big)\,, \tag{44}$$

$$O_t = C_t + \mathrm{Drop}_{.1}(\mathrm{SwiGLU}_{128}(\mathrm{RMSNorm}_F(C_t)))\,. \tag{45}$$

For attention head a with $d_h = d/A = 16$,

$$Q^{(a)} = \bar{Q}_0 W_Q^{(a)} \in \mathbb{R}^{K_q \times d_h}, \tag{46}$$

$$K_t^{(a)} = \bar{E}_t W_K^{(a)} \in \mathbb{R}^{n_t \times d_h}, \tag{47}$$

$$V_t^{(a)} = \bar{E}_t W_V^{(a)} \in \mathbb{R}^{n_t \times d_h}, \tag{48}$$

where E_t stacks the row embeddings. Scores, weights, and pooled outputs are

$$S_t^{(a)} = \frac{Q^{(a)} K_t^{(a)\top}}{\sqrt{d_h}} + M_t, \qquad S_t^{(a)} \in \mathbb{R}^{K_q \times n_t}, \tag{49}$$

$$A_t^{(a)} = \mathrm{softmax}_j\left(S_t^{(a)}\right), \tag{50}$$

$$O_t^{(a)} = A_t^{(a)} V_t^{(a)} \in \mathbb{R}^{K_q \times d_h}. \tag{51}$$

The $1/\sqrt{d_h}$ factor controls score scale: if independent query and key components have unit variance, their dot product has variance d_h; division restores order-one variance [20]. $M_{t,kj} = 0$ for an observed row and $-\infty$ for padding. Concatenating the four heads and projecting supplies the MHA value in Eq. (45). The resulting four 64-vectors are flattened and reduced by

$$a_t = \mathrm{RMSNorm}(\mathrm{GELU}(W_s\,\mathrm{vec}(O_t) + b_s))\,, \qquad W_s \in \mathbb{R}^{64\times 256}. \tag{52}$$

Permutation-invariance derivation. Let P be any $n_t \times n_t$ row-permutation matrix. Permuting rows changes keys and values to PK and PV. The new score matrix is $Q(PK)^\top = QK^\top P^\top$. Rowwise softmax commutes with column permutation, so

$$\mathrm{softmax}(QK^\top P^\top) = \mathrm{softmax}(QK^\top)P^\top. \tag{53}$$

Multiplying by PV cancels the permutation:

$$\mathrm{softmax}(QK^\top)P^\top PV = \mathrm{softmax}(QK^\top)V. \tag{54}$$

Therefore query pooling is invariant to row order, provided the features and mask receive the same permutation. The extractor still stores rows canonically for deterministic serialization.

The attention score term is $O(K_q n_t d_h)$ per head, or $O(K_q n_t d)$ across heads, instead of the $O(n_t^2 d)$ score term of row self-attention. Linear projections still cost $O(n_t d^2)$. The four queries provide four learned summary slots without assigning a fixed market semantic to any slot.

6.3 Gated row/aggregate fusion

Project the 23 bar features to $b_t \in \mathbb{R}^d$ and concatenate $z_t = [a_t; b_t] \in \mathbb{R}^{2d}$. Fusion is

$$g_t = \mathrm{sigmoid}(W_g z_t + b_g) \in (0,1)^d, \tag{55}$$

$$u_t = g_t \odot a_t + (1 - g_t) \odot b_t + W_r z_t + b_r. \tag{56}$$

The bar projection is $23 \to 64 \to 64$ with GELU and RMSNorm. In Eq. (56), the first two terms interpolate coordinatewise, while $W_r z_t + b_r$ provides a direct learned residual path from both summaries.

6.4 Causally masked temporal encoder

Add row i of a learned 64×64 time-embedding table to each fused token, producing $H_i^{(0)} = u_{t-L+i} + p_i$ for $i = 1, \ldots, L$. In each pre-normalized temporal block, masked self-attention and a gated feed-forward network are applied with residual LayerScale:

$$Z = H + \gamma_A \odot \mathrm{Drop}_{.1}\big(\mathrm{MHA}(\mathrm{RMSNorm}(H); M^{causal}, M_B)\big)\,, \tag{57}$$

$$H' = Z + \gamma_F \odot \mathrm{Drop}_{.1}(\mathrm{SwiGLU}_{192}(\mathrm{RMSNorm}(Z)))\,. \tag{58}$$

The learned vectors $\gamma_A, \gamma_F \in \mathbb{R}^d$ start at 10^{-2}, limiting the initial residual branch magnitude [19]. For feed-forward width $d_f = 3d$, SwiGLU is

$$\mathrm{SwiGLU}(x) = W_o\,[\mathrm{SiLU}(W_g x + b_g) \odot (W_u x + b_u)] + b_o, \tag{59}$$

where $\mathrm{SiLU}(z) = z\,\mathrm{sigmoid}(z)$ [18]. M_B excludes padded history tokens as attention keys. After three blocks, a final RMSNorm produces memory $\mathcal{H} \in \mathbb{R}^{B\times 64\times 64}$.

For temporal positions $i, j \in \{1, \ldots, L\}$,

$$M_{ij}^{causal} = \begin{cases} 0, & j \le i, \\ -\infty, & j > i. \end{cases} \tag{60}$$

At position i, every future logit is $-\infty$; its exponential is zero, so temporal attention reads only positions $j \le i$. The common price anchor and volume reference in Eqs. (23)–(37) are computed at the final cutoff from the full 64-bar context, so an early token can indirectly encode information about later *historical* context bars through those shared coordinates. Every anchor, reference, and token is nevertheless available at the forecast cutoff.

6.5 Parallel horizon decoder

For each horizon h, initialize a decoder token from two learned vectors,

$$D_h^{(0)} = q_h^{hor} + e_h^{hor}. \tag{61}$$

Both learned tables have shape 8×64; their sum initializes the eight tokens rather than representing two separately supervised concepts. Decoder block $\ell \in \{1, 2\}$ first applies unmasked self-attention among the eight horizon tokens, then cross-attention to the history, and then a width-128 gated feed-forward sublayer:

$$A^{(\ell)} = D^{(\ell-1)} + \mathrm{Drop}_{.1}\Big[\mathrm{MHA}\Big(\mathrm{RMSNorm}_S(D^{(\ell-1)})\Big)\Big], \tag{62}$$

$$C^{(\ell)} = A^{(\ell)} + \mathrm{Drop}_{.1}\Big[\mathrm{MHA}\Big(\mathrm{RMSNorm}_Q(A^{(\ell)}), \mathrm{RMSNorm}_M(\mathcal{H}), \mathrm{RMSNorm}_M(\mathcal{H}); M_B\Big)\Big], \tag{63}$$

$$D^{(\ell)} = C^{(\ell)} + \mathrm{Drop}_{.1}\Big[\mathrm{SwiGLU}_{128}(\mathrm{RMSNorm}_F(C^{(\ell)}))\Big]. \tag{64}$$

The self-attention is unmasked because no token contains a realized target or earlier forecast. The cross-attention uses four heads of width 16, with the decoder tokens as queries and the 64 historical states as keys and values. Thus all eight outputs are direct functions of the same cutoff history.

6.6 Output projection and in-graph constraints

Final decoder normalization followed by one affine projection gives

$$R = \mathrm{reshape}_{8\times7\times3}\Big(W_o\,\mathrm{RMSNorm}_O(D^{(2)}) + b_o\Big), \qquad W_o \in \mathbb{R}^{21\times64}. \tag{65}$$

The last axis of R is (c, s^-, s^+). Equations (69) and (75), plus tanh for delta, execute inside the forward graph before either the training loss or ONNX output is produced. The resulting $Q \in \mathbb{R}^{B\times8\times7\times3}$ is therefore the constrained tensor consumed by both training and inference.

6.7 Complexity and tensor accounting

With average row count $\bar{n}$, temporal depth $N_T = 3$, decoder depth $N_D = 2$, and fixed model width, the principal attention-score complexity is

$$O(LK_q\bar{n}d) + O(N_T L^2 d) + O(N_D HLd) + O(N_D H^2 d). \tag{66}$$

The four terms are row-query pooling, temporal self-attention, history cross-attention, and horizon self-attention. With $L = 64$ and $H = 8$, one temporal layer has only $64^2 = 4{,}096$ position pairs per head. Peak row activation storage is proportional to the padded batch width rather than a global 1,024-row rectangle for every bar. The full shape trace and parameter subtotals appear in Eq. (2) and Table 2.

7 Outputs, physical constraints, and training objective

7.1 A noncrossing marginal parameterization

For each horizon and target, the unconstrained head emits a center c and two spread logits s^-, s^+. Define

$$\widetilde{Q}_{0.10} = c - \mathrm{softplus}(s^-) - \epsilon_q, \tag{67}$$

$$\widetilde{Q}_{0.50} = c, \tag{68}$$

$$\widetilde{Q}_{0.90} = c + \mathrm{softplus}(s^+) + \epsilon_q, \tag{69}$$

where $\mathrm{softplus}(s) = \log(1 + e^s) > 0$ and $\epsilon_q = 10^{-4}$. Therefore

$$\widetilde{Q}_{0.10} < \widetilde{Q}_{0.50} < \widetilde{Q}_{0.90} \tag{70}$$

for every finite pair of spread logits. Differentiating gives $d\,\mathrm{softplus}(s)/ds = \mathrm{sigmoid}(s)$, so spread gradients are smooth and bounded between 0 and 1. This guarantees order; it does not guarantee that empirical exceedance frequencies equal 10%, 50%, and 90%.

7.2 Quantilewise price constraints

All five price heads use the same anchor-relative coordinate. For a fixed τ, let $\widetilde{L}_\tau, \widetilde{H}_\tau, \widetilde{P}_\tau, \widetilde{B}_\tau, \widetilde{S}_\tau$ denote unconstrained low, high, total-POC, buy-POC, and sell-POC quantiles after Eq. (69). The implemented map is

$$P_\tau = \widetilde{P}_\tau, \tag{71}$$

$$L_\tau = \min(\widetilde{L}_\tau, P_\tau), \tag{72}$$

$$H_\tau = \max(\widetilde{H}_\tau, P_\tau), \tag{73}$$

$$B_\tau = \min\Big(\max(\widetilde{B}_\tau, L_\tau), H_\tau\Big), \tag{74}$$

$$S_\tau = \min\Big(\max(\widetilde{S}_\tau, L_\tau), H_\tau\Big). \tag{75}$$

It follows immediately that

$$L_\tau \le P_\tau, B_\tau, S_\tau \le H_\tau. \tag{76}$$

Low and high expand only enough to include total POC; side POCs are clamped and do not expand the range. The scalar functions min and max are coordinatewise monotone. Since every input sequence in τ is ordered by Eq. (69), compositions in Eq. (75) preserve within-target order across τ. Delta-fraction quantiles pass through the monotone map tanh, which preserves order while bounding them to $[-1, 1]$.

These relationships are necessary physical conditions, not a full probabilistic model. Quantilewise constraints do not specify cross-target dependence, and $L_{0.10}$ together with $H_{0.90}$ is not a guaranteed joint-candle interval.

7.3 Pinball loss and its quantile derivation

For observation y, forecast q, residual $e = y - q$, and quantile level $\tau \in (0, 1)$, pinball loss is

$$\rho_\tau(e) = \max\{\tau e, (\tau - 1)e\} = \begin{cases} \tau(y - q), & y \ge q, \\ (1 - \tau)(q - y), & y < q. \end{cases} \tag{77}$$

Thus underpredicting q90 costs 0.9 per unit while overpredicting it costs 0.1 per unit. To see why the optimum is a conditional quantile, take expectation under conditional cumulative distribution F. At a continuity point,

$$R(q) = \mathbb{E}[\rho_\tau(Y - q)] \tag{78}$$

$$= (1 - \tau) \int_{-\infty}^{q} (q - y) dF(y) + \tau \int_{q}^{\infty} (y - q) dF(y), \tag{79}$$

$$\frac{dR(q)}{dq} = (1 - \tau)F(q) - \tau[1 - F(q)] = F(q) - \tau. \tag{80}$$

Any stationary minimum therefore satisfies $F(q) = \tau$. With atoms, the subgradient condition is $F(q^-) \le \tau \le F(q)$, which is precisely the definition of a valid τ-quantile [8].

For batch item b, horizon h, target k, and level τ, let $m_{bhk} \in \{0, 1\}$ indicate a valid label. The implemented objective is

$$\mathcal{L} = \frac{\sum_{b,h,k,\tau} m_{bhk} w_h w_k \rho_\tau(y_{bhk} - \widehat{q}_{bhk\tau})}{\sum_{b,h,k,\tau} m_{bhk} w_h w_k}, \qquad w_h = \frac{1}{\sqrt{h}}. \tag{81}$$

The target weights in schema order are $(1.0, 0.7, 0.7, 0.8, 0.8, 0.5, 0.7)$. Normalizing by the sum of active weights prevents a batch with more missing side-POC labels from changing the loss scale solely through its denominator. The $1/\sqrt{h}$ schedule gives near horizons more optimization weight without the steeper $1/h$ decay. Reported cell metrics remain unweighted, making every distant-horizon value directly visible.

8 Training and evaluation design

8.1 Optimization and selection

Training uses AdamW [14], peak learning rate 3×10^{-4}, weight decay 10^{-3}, gradient clipping, linear warm-up followed by cosine decay, deterministic algorithms, and early stopping with patience eight. The nominal batch-item cap is 128; a separate padded-row budget reduces the item count around wide windows without deleting rows. Seeds 17, 29, and 43 use identical data, configuration, split indices, and parameter count.

Table 5: Recorded training configuration shared by all three seeds.

Property	Value
Optimizer / peak learning rate	AdamW / 3×10^{-4}
Weight decay / global gradient-norm clip	10^{-3} / 1.0
Schedule	8% linear warm-up, then cosine decay
Epoch cap / early-stopping patience	30 / 8
Nominal item cap / row-token reference	128 / 64
Dropout / workers	0.10 / 0
Seeds	17, 29, 43

Table 5 records the shared optimizer, schedule, batching, and seed settings. Within each seed, the checkpoint with minimum validation weighted pinball loss is retained. The primary run is the seed with minimum validation loss, with seed number as the tie-breaker. No test score, baseline result, figure, parity measurement, or latency measurement enters selection. This is development selection, not an untouched confirmatory protocol, because the study has no separate outer holdout.

8.2 Causal point baselines

Persistence carries the latest observed bar state to every horizon. In target order its normalized forecast is

$$\widehat{y}_h^{pers} = \left(0, \bar{L}_t, \bar{H}_t, \bar{P}_t^{+}, \bar{P}_t^{-}, f_{t,11}, \mathrm{clip}(\Delta_t/V_t, -1, 1)\right), \qquad h = 1, \ldots, 8, \tag{82}$$

where the four barred price terms are the latest low, high, buy-POC, and sell-POC offsets from the latest total POC in target coordinates, and $f_{t,11}$ is the latest encoded relative-log-volume feature. Only the future total-POC offset is fixed at zero. Persistence has no fitted parameters.

Direct ridge receives only the encoded 23-dimensional bar aggregates. For a 64-bar context $b_1, \ldots, b_L$, its 69-vector is

$$z = \left[b_L;\ \frac{1}{L}\sum_{r=1}^{L} b_r;\ b_L - b_1 \right] \in \mathbb{R}^{69}. \tag{83}$$

Thus the "trend" block is an endpoint difference, not a fitted slope. On training windows only, each feature has mean μ_j and population standard deviation

$$s_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(z_{ij} - \mu_j)^2}; \qquad s_j \leftarrow 1 \text{ if } s_j < 10^{-8}. \tag{84}$$

Append an unpenalized intercept to obtain $\widetilde{x}_i = [(z_i - \mu) \oslash s; 1]$ and design matrix $\widetilde{X}$. For each horizon-target cell, only valid training labels are retained. With $\alpha = 1$ and $R = \mathrm{diag}(\alpha, \ldots, \alpha, 0)$, the fitted coefficients solve

$$(\widetilde{X}^\top \widetilde{X} + R)\widehat{\beta} = \widetilde{X}^\top y. \tag{85}$$

The linear system is solved directly. Inference uses the frozen training mean and scale. All non-delta outputs are clipped to $[-2, 2]$ and delta to $[-1, 1]$. Ridge and persistence use the same test indices as FDT-N.

These controls compare FDT-N's q50 point with simple causal point forecasts; only FDT-N produces q10 and q90 in this experiment.

8.3 Metrics

For a horizon-target cell with N valid examples, realization y_i, and FDT-N quantiles $q_{i,.1}, q_{i,.5}, q_{i,.9}$, define

$$\mathrm{MAE}_{.5} = \frac{1}{N}\sum_{i=1}^{N} |y_i - q_{i,.5}|, \tag{86}$$

$$\mathrm{Bias}_{.5} = \frac{1}{N}\sum_{i=1}^{N} (q_{i,.5} - y_i), \tag{87}$$

$$\mathrm{Coverage}_{.8} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[q_{i,.1} \le y_i \le q_{i,.9}], \tag{88}$$

$$\mathrm{Width}_{.8} = \frac{1}{N}\sum_{i=1}^{N} (q_{i,.9} - q_{i,.1}). \tag{89}$$

Pinball loss is reported separately at each quantile using Eq. (77). Coverage and width are complementary: pooled width appears with the selected-seed summary, while Figure 5 shows target-level coverage. POC direction agreement is

$$\frac{1}{N}\sum_i \mathbf{1}[\mathrm{sign}(q_{i,.5}) = \mathrm{sign}(y_i)], \tag{90}$$

a secondary anchor-offset diagnostic, not a trading-return statistic. The legacy artifact field `median_mae_normalized` means MAE of the q50 forecast in Eq. (86); it never means the median of the absolute-error sample.

8.4 Paired contiguous-block resampling

For a fixed target-horizon cell and comparator c, let $m_i = 1$ only when the target is valid for both forecasts. Define the paired absolute-loss difference on that common support as

$$d_i = |y_i - q_{i,.5}^{FDT}| - |y_i - \widehat{y}_i^{c}| \qquad \text{for } m_i = 1. \tag{91}$$

$d_i < 0$ indicates lower FDT-N absolute error, and $d_i > 0$ indicates lower comparator error. Ordered test row i receives block $\lfloor (i-1)/240 \rfloor$, producing 13 contiguous blocks: 12 blocks of 240 examples and a final remainder. For resample r, sample 13 block indices with replacement using the run seed and compute the paired-valid observation-weighted mean

$$\overline{d}^{*(r)} = \frac{\sum_{b\in I_r}\sum_{i\in\mathcal{B}_b} m_i d_i}{\sum_{b\in I_r}\sum_{i\in\mathcal{B}_b} m_i}, \qquad r = 1,\ldots,2000, \tag{92}$$

The denominator is the resampled paired-valid count and must be positive; missing side-POC labels are excluded rather than treated as zero-error observations. The 2.5th and 97.5th empirical percentiles form a descriptive 95% sensitivity range. Each range is classified as below zero, above zero, or spanning zero.

Four hours exceeds the 72-minute input-plus-label span, but adjacent windows at block boundaries still overlap. Only 13 blocks are available, the block length was not justified by a full dependence study, and no family-wise multiplicity adjustment is applied across 56 cells. The ranges are therefore not independent-block confidence intervals, significance tests, or generalization bounds. No pooled effect interval is computed.

9 Results

9.1 Selection and seed stability

Table 6: All declared seeds. Loss is the weighted pinball objective; MAE is the pooled normalized MAE of the q50 forecast. Seed 17 is selected using validation loss only.

Seed	Epoch	Val. loss	Test loss	q50 MAE	80% coverage	POC sign
17	21	0.112634	0.070392	0.235870	78.98%	54.90%
29	13	0.114391	0.071542	0.237238	78.96%	49.44%
43	20	0.113713	0.070793	0.236171	80.23%	50.71%
Mean ± sample SD	–	0.113579 ± 0.000886	0.070909 ± 0.000584	0.236426 ± 0.000719	$79.39\% \pm 0.73$ pp	$51.69\% \pm 2.86$ pp

Table 6 reports all declared runs. For seed 17, pooled q10, q50, and q90 pinball losses are 0.055493, 0.117935, and 0.051387. Mean central-80% width is 0.720113 and signed q50 bias is +0.034807 in normalized target coordinates. The three runs summarize optimization variation on the fixed corpus and split.

9.2 Comparison by target

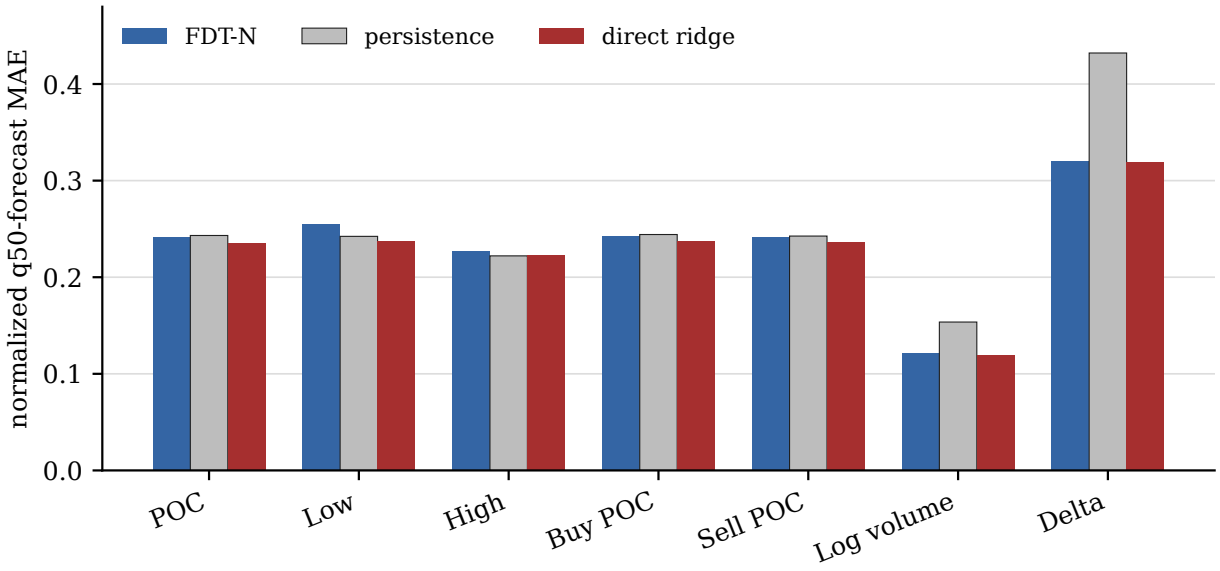


Figure 3: Selected-run normalized q50-forecast MAE by target. Lower is better. FDT-N's persistence improvement is concentrated in flow variables, while direct ridge is lower for every target aggregate.

Figure 3 provides the visual comparison; Table 12 in the appendix gives the exact values and FDT-N marginal coverage.

Pooled FDT-N MAE is 7.25% lower than persistence and 2.68% higher than ridge. Relative log volume and delta fraction account for the aggregate persistence difference; across the other 40 horizon-target cells, FDT-N MAE is higher by 0.002826 on average. Ridge has lower MAE in 55 of 56 individual cells.

9.3 Comparison by horizon

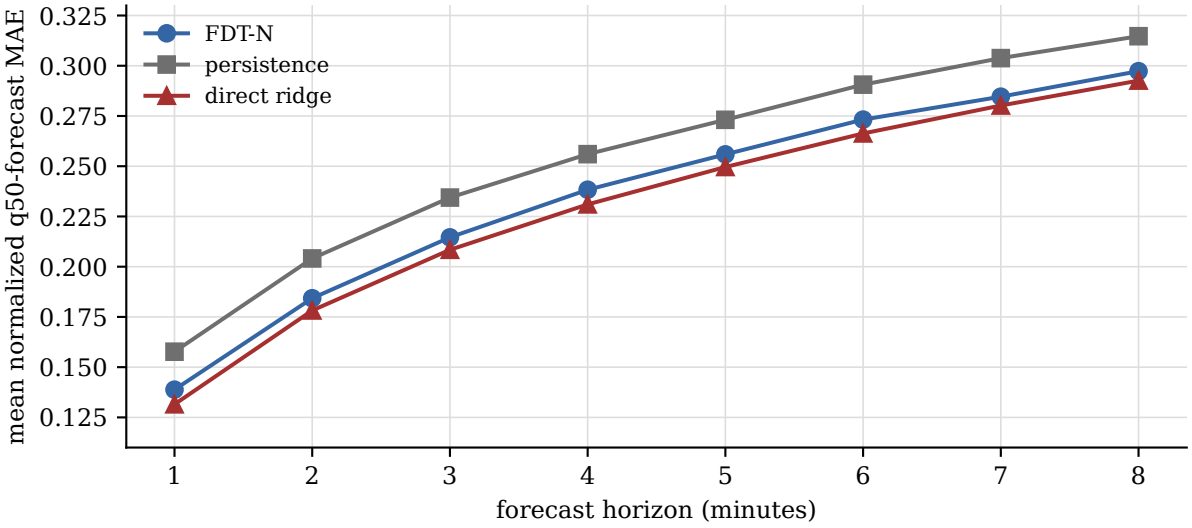


Figure 4: Q50-forecast MAE averaged over seven targets at each horizon. FDT-N lies below persistence and above direct ridge at every horizon aggregate.

Figure 4 shows the horizon profile; Table 13 in the appendix lists the exact one-minute-step aggregates.

Error increases monotonically with horizon for all three methods at the aggregate level. That pattern is descriptive; the overlapping direct targets prevent treating the eight rows as independent replications.

9.4 Marginal coverage

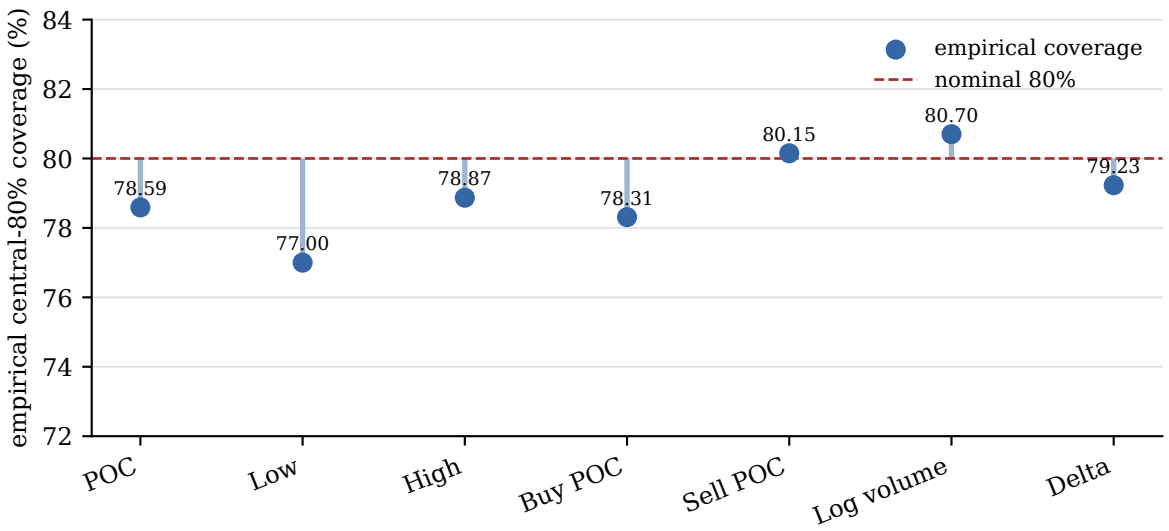


Figure 5: Selected-run empirical central-80% marginal coverage aggregated by target. Points show coverage and the dashed line marks nominal 80%.

Pooled coverage is 78.98%, target aggregates range from 77.00% to 80.70%, and the 56 individual horizon-target cells range from 74.75% to 85.66%. These values are empirical marginal coverage on the chronological test partition; the data split has no separate calibration partition.

9.5 Paired resampling summaries

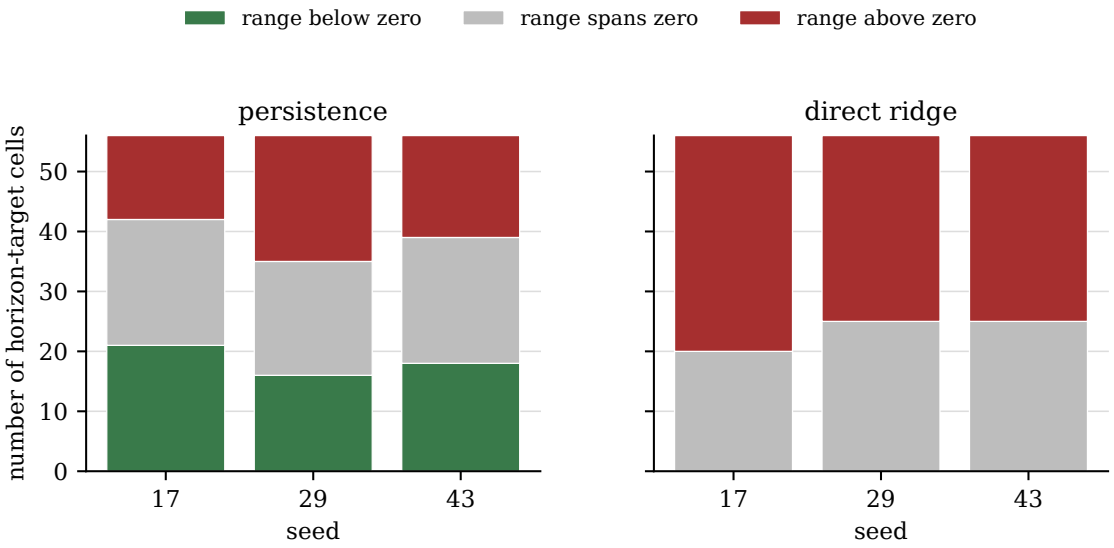


Figure 6: Counts of 56 cellwise descriptive ranges by sign relative to zero, using the definition after Eq. (92).

Figure 6 summarizes the range signs; Table 14 in the appendix lists all seed-comparator counts.

Across all three seeds, the direct-ridge counts are 0 below zero, 20–25 spanning zero, and 31–36 above zero. Section 8.4 defines these ranges as descriptive summaries.

9.6 Clipping, parity, and runtime

Price-target clipping affects approximately 1.04%–1.17% of observations, depending on target and bound. Encoded row-price offset boundary saturation is 3.51%. All reported normalized errors use the stored clipped coordinates.

The validation-selected checkpoint is exported to ONNX. On a representative batch-one window whose widest bar has 187 rows, ONNX Runtime differs from the PyTorch reference by maximum absolute error 2.3842×10^{-7}

and mean absolute error 5.1666×10^{-8}, below the declared 10^{-4} tolerance. Across 50 isolated CPU runs, latency is 11.410/14.641/17.240 ms at p50/p95/p99. These numbers measure isolated batch-one model execution.

10 Interpretation

10.1 End-to-end behavior

The implementation demonstrates that variable-size footprint rows can be represented without silent compaction, pooled by a constant number of queries, combined with engineered bar features, and mapped to ordered multi-horizon marginals in a 371,861-parameter model. The three training runs have similar aggregate test loss and MAE. The selected export has strong numerical agreement across PyTorch and ONNX Runtime, and its pooled central-80% coverage is near 80% on this test partition.

The point comparison has a consistent profile: FDT-N's pooled q50 error is lower than persistence, principally through relative log volume and delta fraction, while direct ridge is lower for every target aggregate, every horizon aggregate, and 55 of 56 individual cells. The paired ridge ranges are either above zero or span zero under the sign convention in Section 8.4.

10.2 Benchmark profile

The latest, mean, and endpoint-difference aggregates contain substantial short-horizon location information in this period. Each ridge cell specializes in one conditional-location estimate, whereas FDT-N distributes 371,861 parameters across three quantiles, seven targets, and eight horizons. The comparison therefore supplies a useful scale reference for the richer output task.

Component attribution is a separate experiment. An aggregate-only FDT-N comparison would quantify the contribution of full rows; mean, max, one-query, and fixed-query controls would quantify pooling; and row-only versus aggregate-only variants would quantify the fusion paths. Query interpretation requires feature permutation, query swaps, interventions, and stability tests rather than attention visualization alone [7].

10.3 Follow-up experiments

A follow-up study can pre-register a later outer period and use anchored walk-forward development folds. On identical windows it can compare: aggregate-only versus full-row input; mean, max, one-query, and four-query pooling; row-only versus aggregate-only versus gated fusion; equal versus $1/\sqrt{h}$ horizon weights; matched-capacity causal GRU or TCN; train-only quantile regression and quantile trees; and a task-adapted time-series transformer where compute and tuning budgets are comparable. A separate chronological calibration partition is needed for any conformal or post-hoc interval adjustment.

Every target-horizon-quantile cell, clipping rate, missing-label count, and conditional slice can be reported with a pre-specified dependence unit, multiplicity policy, and stopping rule. Cross-instrument or cross-venue evaluation requires corresponding datasets and coordinates.

11 Scope of the present study

Data scope. The corpus covers fourteen days, one instrument, one venue, a 2.5-USDT row grid, and a maximum of 605 rows per bar. Trade-side classification, exchange corrections, row snapping, and quantity rounding are part of the stated data-generating procedure.

Selection scope. Seed selection uses validation loss on the single chronological split. Architecture and preprocessing belong to the same development study; an outer-period evaluation is listed as follow-up work.

Comparator scope. Persistence and ridge are q50 point baselines. The reported pinball loss and interval coverage belong to FDT-N; matched distributional comparators and component ablations are follow-up experiments.

Statistical scope. Input windows, target horizons, and adjacent blocks overlap. The 13-block resampling ranges are descriptive, and the three seeds summarize optimization variation on the fixed dataset and split.

Output scope. Pinball loss and coverage evaluate targetwise conditional marginals. POC sign agreement describes the anchor-offset sign; economic or execution outcomes are outside the output contract.

Runtime scope. Parity and latency refer to the frozen batch-one model artifact. Data ingestion, bar construction, chart rendering, and downstream decisions are separate system components.

12 Reproducibility, availability, and declarations

The source manifest binds the 14 official archives, while the processed dataset, three result files, selected checkpoint, ONNX graph, metadata, and parity fixture are bound by SHA-256. Raw archives are recoverable from Binance Data Vision. The submission bundle directly includes the executable ONNX graph, metadata, and artifact-bound input/output fixture listed in Table 7; the complete experiment lineage is recorded in `../research/evidence-ledger.md`.

Table 7: Packaged execution artifacts and identities. Paths are relative to this source directory.

Artifact	Package path	SHA-256
ONNX graph	`../model/fdt-n.onnx`	`adce8cd39312bf3ad1389a23ed5ca24d6f47a7f58dbb26d76ae7d54af1011af7`
Metadata	`../model/metadata.json`	`9f998ac6b56b008ac02aaa0ed5c4f08447b1735f33027294773d9e1aa0867763`
Parity fixture	`../model/parity-fixture.json`	`3af05175fb8c2d23f8dd33eb818c6c94afccc79135364149ef501bcf33f53160`

The recorded environment includes Python 3.12.6, PyTorch 2.5.1, NumPy 2.1.2, ONNX 1.22.0, and ONNX Runtime 1.28.0. Result identities, seeds, configuration, parameter count, best epoch, elapsed time, and device class are preserved in the evidence ledger.

The study uses public market-event archives and contains no personal data or human participants. FDT-N's defined output is the marginal-quantile tensor in Table 1; downstream economic decisions are outside that interface. Dhruv Sharma is the author and performed the study, analysis, artifact preparation, and manuscript writing.

Funding and competing interests. The artifact record identifies no external funding source or competing interest for this study.

13 Conclusion

FDT-N is a compact hierarchical transformer for direct marginal quantile forecasts from ragged volume-footprint rows and engineered bar features. The mathematical construction is causal at the bar cutoff, permutation-invariant within row sets, non-autoregressive across horizons, noncrossing within each target, and constrained by basic price-range relationships. The input contract retains every row for accepted bars, and the selected export passes its numerical parity tolerance.

Across three seeds, pooled normalized q50 MAE is 0.23587–0.23724. For the selected seed it is 0.235870, compared with 0.254298 for persistence and 0.229719 for direct ridge; pooled central-80% marginal coverage is 78.98%. The paper's four-tensor input contract, complete layer equations, parameter accounting, and artifact-bound forward trace make the transform from footprint rows to $[B, 8, 7, 3]$ outputs explicit and reproducible.

A Worked input and output examples

A.1 Footprint arithmetic

Consider the illustrative footprint in Table 8 with row size 0.25 price units. These values demonstrate arithmetic only; they are not observations from the experiment.

Table 8: Illustrative completed footprint bar.

Price	Buy	Sell	Total	Delta	ZeroBid	ZeroAsk
100.50	8	0	8	+8	1	0
100.25	32	6	38	+26	0	0
100.00	20	30	50	−10	0	0
99.75	5	22	27	−17	0	0
99.50	0	7	7	−7	0	1

Summing Eq. (8) gives $V^+ = 65$, $V^- = 65$, $V = 130$, and $\Delta = 0$. The total volumes are $(8, 38, 50, 27, 7)$, so total POC is 100.00. The largest buy volume is 32 at 100.25 and the largest sell volume is 30 at 100.00, giving buy POC 100.25 and sell POC 100.00.

The total centroid is

$$c = \frac{100.50(8) + 100.25(38) + 100.00(50) + 99.75(27) + 99.50(7)}{130} \tag{93}$$

$$= 100.025. \tag{94}$$

Similarly, $c^+ \approx 100.1654$ and $c^- \approx 99.8846$. For illustration, use an adjacent-larger expansion rule for the 70% value-area target $0.70(130) = 91$. Starting at POC volume 50, compare the adjacent 38 and 27; add 38, giving 88. Then compare the next upper 8 with the lower 27; add 27, giving 115. The resulting contiguous value area is 99.75–100.25.

For the 100.25 row, the offset is one row increment and

$$\frac{v^+}{V} = \frac{32}{130} = 0.2462, \quad \frac{v^-}{V} = \frac{6}{130} = 0.0462, \quad \frac{v}{V} = \frac{38}{130} = 0.2923, \quad \frac{\Delta}{V} = \frac{26}{130} = 0.2000. \tag{95}$$

If the bar POC is 100.00, the first row feature in Eq. (22) is $1/32 = 0.03125$, not the unscaled one-row offset.

Suppose an illustrative one-step q50 predicts normalized POC offset $1/64$, low $-1/64$, high $4/64$, buy POC $1/64$, sell POC 0, normalized relative log volume 0.05, and delta fraction 0.10. The inverse prices are 100.25, 99.75, 101.00, 100.25, and 100.00. If the context reference satisfies $\exp(\widetilde{\ell}_t) = 101$, then

$$\widehat{V} = \exp(\widetilde{\ell}_t + 4(0.05)) - 1 = 101e^{0.2} - 1 \approx 122.36. \tag{96}$$

Multiplying this q50 volume plug-in by the q50 delta fraction gives 12.24, but that product is not the q50 of delta volume because quantiles do not generally commute with products of dependent random variables.

Finally, suppose illustrative POC q10/q50/q90 offsets are $(-1, 1, 3)$ row increments and the realization is 2. By Eq. (77), the losses are

$$0.1(2 - (-1)) = 0.3, \qquad 0.5(2 - 1) = 0.5, \qquad 0.1(3 - 2) = 0.1, \tag{97}$$

whose unweighted mean is 0.3 row increment.

A.2 Artifact-bound forward pass

This example is read directly from `../model/parity-fixture.json` and executed by the packaged `../model/fdt-n.onnx`. The deterministic fixture rule chooses the earliest window whose padded row width reaches the corpus p95, so selection depends on input width. Its last context bar starts at 2024-01-10 19:13:00 UTC and the first forecast cutoff is 19:14:00 UTC. The last-context POC anchor is 46,465 USDT, the row size is 2.5 USDT, and the context log-volume reference is $\widetilde{\ell}_t = 11.59615697$.

Table 9: The four actual fixture inputs. $N = 187$ is the widest bar in this window; valid row counts range from 8 to 187 and padding is identified only by the boolean mask.

Tensor	Shape	Content
`row_features`	$[1, 64, 187, 11]$	encoded rows, oldest bar to newest bar
`row_mask`	$[1, 64, 187]$	observed-row indicators; last bar has 56 rows
`bar_features`	$[1, 64, 23]$	aggregate history in schema order
`bar_mask`	$[1, 64]$	64 valid completed bars

Table 9 makes the dynamic row width and both masks concrete for this window. Table 10 records actual intermediate tensors from that pass, restoring the batch axis to the row-set summary. "First four" means the first four values in deterministic flattened storage; the statistic is RMS except for the sigmoid gate, for which the mean is more interpretable.

Table 10: Artifact-bound processing trace.

Stage	Shape	Statistic	First four values
Row-set summary A	$[1, 64, 64]$	0.982983	0.818572, 1.133166, −0.201165, −0.125622
Bar summary	$[1, 64, 64]$	1.008219	−0.568567, 0.931088, −0.373822, −2.358358
Fusion gate	$[1, 64, 64]$	0.527938	0.508116, 0.125989, 0.452889, 0.394906
Fused history U	$[1, 64, 64]$	1.013353	0.188924, 1.828844, 0.049782, −1.720833
Temporal memory $\mathcal{H}$	$[1, 64, 64]$	1.004026	0.224272, 2.108123, −0.058805, −1.318258
Decoder $D^{(2)}$	$[1, 8, 64]$	1.991881	−1.489616, −0.704202, 0.551906, 2.596395
Affine head R_{flat} (pre-reshape)	$[1, 8, 21]$	0.838586	−0.191553, −0.529968, −0.584842, 0.546425

For horizon one, Table 11 shows the raw head triples and the actual ONNX return. For example, POC uses

$$q_{.1} = -0.191553 - \text{softplus}(-0.529968) - 10^{-4} = -0.654521, \tag{98}$$

with q50 equal to the center and q90 formed by the upper softplus spread. Low and side-POC values are then clamped by Eq. (75); delta values pass through tanh.

Table 11: Actual horizon-one raw head and constrained normalized output.

	Raw (c, s^-, s^+)			Returned (q10, q50, q90)		
Target	c	s^-	s^+	q10	q50	q90
POC	−0.191553	−0.529968	−0.584842	−0.654521	−0.191553	0.251432
Low	0.546425	−1.126999	0.763642	−0.654521	−0.191553	0.251432
High	0.054772	−1.039371	−0.635295	−0.248153	0.054772	0.479995
Buy POC	−0.810497	0.995014	−0.557284	−0.654521	−0.191553	0.251432
Sell POC	−0.887576	1.075034	0.601971	−0.654521	−0.191553	0.251432
Log volume	−0.148085	−2.036494	−1.665860	−0.270832	−0.148085	0.025151
Delta fraction	0.050435	−1.225202	−1.040504	−0.204256	0.050392	0.339090

The price inverse is $46{,}465 + 2.5(64)q$, and scaled volume is divided by the recorded quantity scale 1,000 to obtain BTC. The returned physical marginals are therefore

Target	q10	q50	q90
POC / low / buy POC / sell POC (USDT)	46,360.28	46,434.35	46,505.23
High (USDT)	46,425.30	46,473.76	46,541.80
Volume (BTC)	36.783	60.102	120.181
Delta fraction	−0.204256	0.050392	0.339090

PyTorch and ONNX Runtime return the same full $[1, 8, 7, 3]$ tensor within maximum absolute difference 2.3842×10^{-7} and mean absolute difference 5.1666×10^{-8}.

B Detailed benchmark tables

Table 12: Selected-run target aggregates. Coverage is empirical marginal q10–q90 coverage for FDT-N; point baselines have no interval.

Target	FDT-N MAE	Persistence MAE	Ridge MAE	FDT-N coverage
POC offset	0.2414	0.2432	0.2357	78.59%
Low offset	0.2552	0.2423	0.2371	77.00%
High offset	0.2272	0.2221	0.2225	78.87%
Buy-POC offset	0.2431	0.2442	0.2374	78.31%
Sell-POC offset	0.2415	0.2426	0.2359	80.15%
Relative log volume	0.1219	0.1536	0.1198	80.70%
Delta fraction	0.3207	0.4321	0.3197	79.23%
Pooled	0.235870	0.254298	0.229719	78.98%

Table 13: Selected-run horizon aggregates. Horizons are one-minute steps.

Horizon	FDT-N MAE	Persistence MAE	Ridge MAE	FDT-N coverage
1	0.1388	0.1577	0.1315	83.41%
2	0.1843	0.2041	0.1782	79.45%
3	0.2146	0.2344	0.2084	78.55%
4	0.2383	0.2560	0.2310	78.80%
5	0.2559	0.2731	0.2496	77.44%
6	0.2732	0.2906	0.2663	77.64%
7	0.2846	0.3038	0.2802	78.07%
8	0.2973	0.3147	0.2926	78.45%

Table 14: Cell counts under paired four-hour block resampling.

Seed	Comparator	Below zero	Spans zero	Above zero
17	Persistence	21	21	14
29	Persistence	16	19	21
43	Persistence	18	21	17
17	Direct ridge	0	20	36
29	Direct ridge	0	25	31
43	Direct ridge	0	25	31

C Notation and dimensional reference

Table 15 collects the symbols used throughout the derivation.

Table 15: Principal notation.

Symbol	Meaning
t, j, h, k, τ	bar, row, forecast horizon, target, and quantile indices
L, H, K, Q	64 history bars, 8 horizons, 7 targets, and 3 quantile levels
p, v^+, v^-, v, Δ	row price, buy volume, sell volume, total volume, and signed delta
R_t, n_t	complete accepted row set for bar t and its variable cardinality
$B_t = (R_t, b_t)$	completed footprint bar with row set and 23-vector aggregates
δ_p, S_g, S_p, S_v	2.5-USDT row size; 32-row input-gap scale; 64-row target scale; log-volume scale 4
d, A, d_h, K_q	model width 64, 4 heads, head width 16, and 4 row queries
m_{bhk}	target-validity mask; padding has a separate row-attention mask
$q_{i,.1}, q_{i,.5}, q_{i,.9}$	predicted marginal q10, q50, and q90 in normalized target coordinates
ρ_τ	pinball loss at quantile level τ

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